

Method

We are trying to check if we can map the edges in the tree to pairwise disjoint paths (no same edges) in the network with the appropriate starting and ending points.

If there are C edges in the tree let us color the edges in the network with $0, 1, \dots, C$ colors

Color $k = 0, 1, \dots, C - 1$ designates that edge e is in the path that corresponds with edge k in the tree

Color $k = C$ is neutral color, it means that the edge is not taken to account

Continuation constraint:

- An edge can have color $k = 0, 1, \dots, C - 1$ if its tail is the associated tail in the tree and no other edges point to that node or it is not and exactly one edge points there
It is the same with the head
- Hence an edge can be colored if either it is an edge from a simple path from the tail node from the tree to the head node from the tree or it is part of a directed cycle
- Since the network is DAG it has to be the path

Start One constraint

- Since we want every edge in the tree to have an associated path we put a constraint that if edges a starts from the tree node there has to be exactly one edge colored a in the network thence exactly one path
- It should be enough to only put a constraint on the tail nodes but we also putted on the heads for symmetry
- We added dummy colors and edges for every node so we can always put two inputs to the constraint

Since we add the StartOne constraint to every tail node in the tree we ensure that every edge in the tree has an associated colored path, and because we add the Continuation constraint to every edge we ensure that that path is unique since it cannot diverge at any point. There is an associate coloring, that is a contained tree in the network, iff the model can sample otherwise we raise error.